

G -theory of group rings for finite groups

Julia Semikina
University of Lille

§ 1 K-theory

$\mathcal{C}$ -exact category



category QE

$$\rightsquigarrow K_i(\mathcal{C}) = \pi_i(BQE)$$

Quillen's
 Q -construction

① $\text{Proj } R = \text{f.g. projective } R\text{-mod}$
 $K(R) := K(\text{Proj } R)$

Topological invariants

Wall's finiteness obstr. $\rightsquigarrow K_0(\mathbb{Z}\pi)$

Whitehead torsion $\rightsquigarrow K_1(\mathbb{Z}\pi)$

Hard to compute

② $\text{Mod } R = \text{f.g. } R\text{-modules}$
 $G(R) = K(\text{Mod } R)$

Dynamical systems invariants

Lefschetz invariant
(existence of Morse-Smale diffeo)

$$\rightsquigarrow G(\mathbb{Z}^n)$$

Even harder

§2 Decomposition formula for $G(\mathbb{Z}\Gamma)$

Let Γ be a finite abelian group

Proved

Lenstra
& Webb

$$G_n(\mathbb{Z}\Gamma) \cong \bigoplus_{C \in \underbrace{C(\Gamma)}_{\text{set of cyclic quotients of } \Gamma}} G_n(\mathbb{Z}[\xi_{|C|}, \frac{1}{|C|}])$$

$$\mathbb{Q}\Gamma \cong \bigoplus_{\substack{\rho \in X(\Gamma) \\ \text{irr. rational reps of } \Gamma}} M_{n_\rho}(\mathcal{D}_\rho^{\text{op}}) \quad \left[\begin{array}{l} \rho: \Gamma \rightarrow \text{Aut}(V_\rho) \\ \mathcal{D}_\rho = \text{End}_{\mathbb{Q}\Gamma}(V_\rho) \end{array} \right] \rightsquigarrow \Lambda_\rho = \text{maximal } \mathbb{Z}\text{-order in } \mathcal{D}_\rho$$

Conjectured

Hambleton
Taylor
Williams

$$G_n(\mathbb{Z}\Gamma) \cong \bigoplus_{\rho \in X(\Gamma)} G_n(\Lambda_\rho[\frac{1}{\omega_\rho}])$$

$$\omega_\rho = \frac{|\Gamma|}{d_\rho k_\rho}$$

$$k_\rho = |\ker(\rho)|$$

$d_\rho = \dim$ of irr.
complex const. $\bigotimes_{\mathbb{Q}} \mathbb{C}$

$$G_n(\mathbb{Z}\Gamma) \cong \bigoplus_{\mathfrak{p} \in X(\Gamma)} G_n(\Lambda_{\mathfrak{p}}[\frac{1}{w_{\mathfrak{p}}}])$$

Shown for:

- abelian groups

[Lenstra '81, Webb '86]

- nilpotent groups

[Hambleton-Taylor-Williams '88]

- groups of square-free order

[Webb '87]

Fails for:

- S_5 [Webb-Yao '93]

simple

- $SL(2, \mathbb{F}_3)$ [S '2018]

solvable

§ 3 Comparison of ranks for G_1

Keying: computation of $\text{rk } G_1(\mathbb{Z}\Gamma)$

$$\text{rk } G_1(\mathbb{Z}\Gamma) = \sum_{\rho \in X(\Gamma)} \text{rk } G_1(\wedge_{\rho} [\frac{1}{\omega_{\rho}}])$$



⊛ $\# \{ \text{irr. } \mathbb{F}_p\text{-rep. of } \Gamma \} = \sum_{\rho \in I_p} t_{\rho}$

irr. rational rep.
for which ω_{ρ}
is not divisible by p

t_{ρ} ← # prime ideals
in $\mathcal{O}_{\rho}$
that divide (p) ,

where $\mathcal{O}_{\rho}$ is the ring
of integers of
 $E_{\rho} = \mathbb{Q}(\chi_{\rho})$

with χ_{ρ} = character of
irr. const. of $\mathbb{C} \otimes_{\mathbb{Q}} \mathbb{F}$.



False:

not every $\mathbb{F}_p$ -rep.
is obtained by reduction mod p
of a defect zero block.



$\geq$ holds